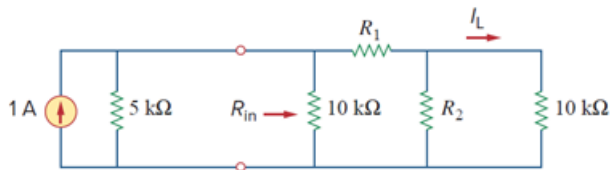


Problem

For a specific application, the circuit shown in Fig. 2.140 was designed so that $I_L = 83.33 \text{ mA}$ and that $R_{in} = 5 \text{ k}\Omega$. What are the values of R_1 and R_2 ?

Figure 2.140:



Step-by-step solution

Step 1 of 4

Refer to the circuit in Figure 2.140 in the textbook.

Input resistance is $R_{in} = 5 \text{ k}\Omega$, load current is $I_L = 83.33 \text{ mA}$.

But from the figure it is clear that R_{in} is calculated as,

$$\begin{aligned} R_{in} &= 10 \text{ k}\Omega \parallel \left[R_1 + (R_2 \parallel 10 \text{ k}\Omega) \right] \\ &= 10 \text{ k}\Omega \parallel \left[R_1 + \left(\frac{R_2 \times 10 \text{ k}\Omega}{R_2 + 10 \text{ k}\Omega} \right) \right] \\ &= 10^4 \parallel \left[\frac{R_1 (R_2 + 10^4) + (10^4 R_2)}{(R_2 + 10^4)} \right] \end{aligned}$$

Step 2 of 4

Substitute $5 \text{ k}\Omega$ for R_{in} .

$$\begin{aligned} 5 \times 10^3 &= \frac{10^4 \left[R_1 (R_2 + 10^4) + 10^4 R_2 \right]}{10^4 (R_2 + 10^4) + (R_1 (R_2 + 10^4) + 10^4 R_2)} \\ 10^4 (R_2 + 10^4) + (R_1 (R_2 + 10^4) + 10^4 R_2) &= 2 \left[R_1 (R_2 + 10^4) + 10^4 R_2 \right] \\ 10^4 (R_2 + 10^4) &= R_1 (R_2 + 10^4) + 10^4 R_2 \\ R_1 (R_2 + 10^4) &= 10^8 \quad \dots\dots (1) \\ R_1 &= \frac{10^8}{(R_2 + 10^4)} \end{aligned}$$

Voltage across resistor R_2 is,

$$\begin{aligned} V_{R_2} &= I_L \times 10 \times 10^3 \\ &= 83.33 \times 10^{-3} \times 10^4 \\ &= 833.3 \text{ V} \end{aligned}$$

Step 3 of 4

As $R_{in} = 5 \text{ k}\Omega$ and $5 \text{ k}\Omega$ resistor are in parallel the current division takes place equal amounts of current flows through each resistor.

Voltage across $5 \text{ k}\Omega$ resistor is given by

$$0.5 \times 5 \times 10^3 = 2500 \text{ V}$$

But by applying Kirchhoff's current law we can write

$$\begin{aligned} \frac{2500 - V_{R_2}}{R_1} &= \frac{V_{R_2}}{R_2} + I_L \\ \frac{2500 - 833.3}{R_1} &= \frac{833.3}{R_2} + 83.3 \times 10^{-3} \\ \frac{1666.67}{R_1} - 83.3 \times 10^{-3} &= \frac{833.3}{R_2} \quad \dots\dots\dots (2) \\ \frac{1666.67 - 83.3 \times 10^{-3} R_1}{R_1} &= \frac{833.3}{R_2} \\ R_2 &= \left[\frac{833.3}{1666.67 - 83.3 \times 10^{-3} R_1} \right] \end{aligned}$$

Step 4 of 4

Substitute equation (1) in equation (2) we get,

$$\begin{aligned} R_2 &= \left[\frac{833.3}{1666.67 - 83.3 \times 10^{-3} \frac{10^8}{(R_2 + 10^4)}} \right] \\ &= \left[\frac{833.3(R_2 + 10^4)}{1666.67 - 83.3 \times 10^{-3} \times 10^8} \right] \\ 1666.67 R_2^2 + 833.67 \times 10^4 R_2 - 833.3 R_2 &= 833.3 \times 10^4 \\ 1666.67 R_2^2 + 833.59 \times 10^4 R_2 - 833.3 \times 10^4 &= 0 \end{aligned}$$

On solving the above equation we get $R_2 = 5 \text{ k}\Omega$.

Substitute $R_2 = 5 \text{ k}\Omega$ in equation (1) we get.

$$\begin{aligned} R_1 &= \frac{10^8}{(5 \times 10^3 + 10^4)} \\ &= 6666.67 \Omega \\ &\cong 6.67 \text{ k}\Omega \end{aligned}$$

Thus, the values of unknown resistors are, $R_1 = 6.67 \text{ k}\Omega, R_2 = 5 \text{ k}\Omega$.